SIMATS ENGINEERING

ECA09 - DIGITAL SIGNAL PROCESSING LAB

Experiment 05.3

Design and Analysis of an FIR High-Pass Filter Using the Hanning Window Technique in MATLAB.

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Aim

To implement and analyze the experiment described in the official ECA09 MATLAB lab manual.

Procedure / Calculations

Follow the experiment workflow given in the official lab manual: initialize MATLAB, define the specified signal/filter parameters, execute the algorithm, plot the required responses, and record the result.

MATLAB Program

```
clc; clear; close all;
%% --- Input Parameters ---
fp = 1000;        % High-pass cutoff frequency (Hz)
fs = 8000;        % Sampling frequency (Hz)
N  = 50;          % Filter order
wc = fp/(fs/2);   % Normalized cutoff (0-1)
%% --- FIR High Pass using Hanning Window ---
b_hann = fir1(N, wc, 'high', hanning(N+1));
disp('--- High-Pass Coefficients (Hanning Window) ---');
disp(b_hann');
%% --- Frequency Response Plot ---
figure;
freqz(b_hann,1);
title('High-pass FIR using Hanning Window');
```

Output / Analysis

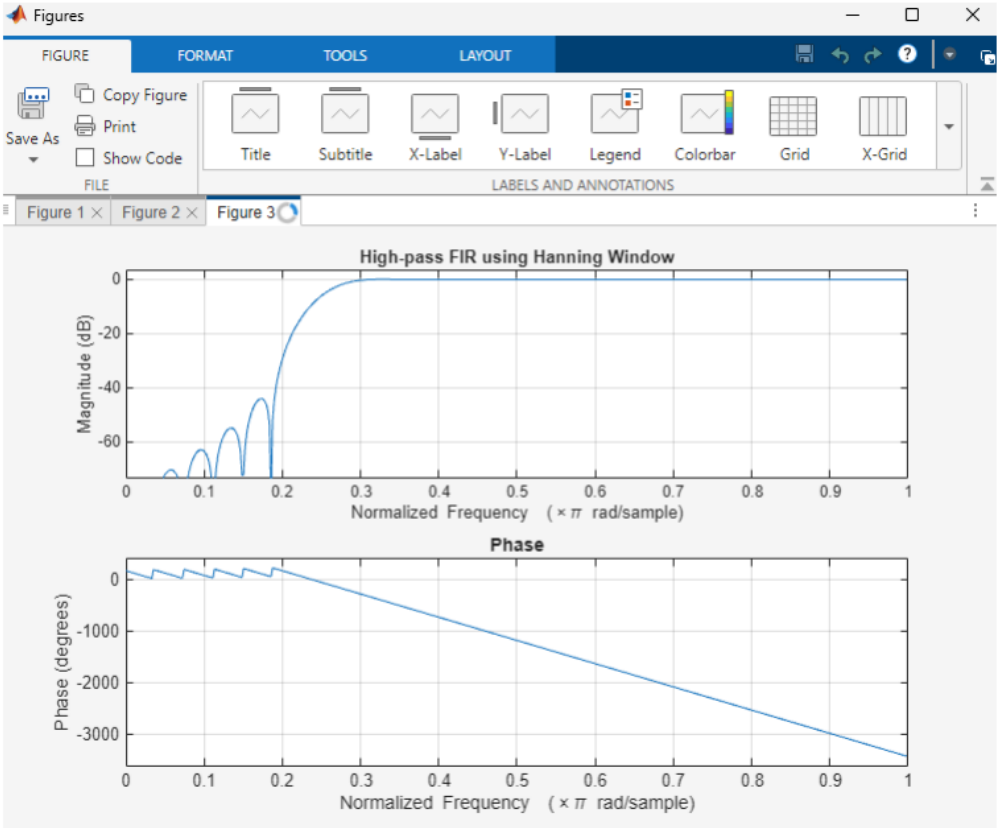
The output/analysis pages below are reproduced from the uploaded official ECA09 MATLAB lab manual for this experiment.

```
b_hann = fir1(N, wc, 'high', hanning(N+1));

disp('--- High-Pass Coefficients (Hanning Window) ---');

disp(b_hann');

%% --- Frequency Response Plots ---

figure;

freqz(b_hann,1);

title('High-pass FIR using Hanning Window');
```

Output:

```
--- High-Pass Coefficients (Hanning Window) ---
  -0.0000
  -0.0000
   0.0003
   0.0008
   0.0009
  -0.0000
  -0.0020
  -0.0038
  -0.0035
  -0.0000
   0.0057
   0.0100
   0.0087
  -0.0000
  -0.0127
  -0.0216
  -0.0183
  -0.0000
   0.0267
   0.0464
   0.0410
  -0.0000
  -0.0726
  -0.1568
  -0.2243
   0.7500
  -0.2243
  -0.1568
  -0.0726
  -0.0000
   0.0410
```

0.0464
0.0267
-0.0000
-0.0183
-0.0216
-0.0127
-0.0000
0.0087
0.0100
0.0057
-0.0000
-0.0035
-0.0038
-0.0020
-0.0000
0.0009
0.0008
0.0003
-0.0000
-0.0000



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Result

1. The FIR High-Pass Filter was successfully designed using the Hanning Window technique for the specified cutoff frequency, sampling frequency, and filter order.

2. The FIR filter coefficients were successfully generated using MATLAB and the designed filter was implemented.

3. The magnitude response and phase response of the designed FIR High-Pass Filter were obtained and analyzed. The magnitude response exhibited attenuation of low-frequency components and passage of high-frequency components, confirming the desired high-pass filtering characteristics. The phase response demonstrated the expected linear-phase behavior of the FIR filter.

Practical Question 1

Question:

Design an FIR High-Pass Filter using the **Hanning (Hann) Window** technique in MATLAB for a sampling frequency of **10000 Hz**, cutoff frequency of **2500 Hz**, and filter order of **50**.

Tasks:

1. Calculate the normalized cutoff frequency.
2. Generate the FIR filter coefficients using the fir1() function with a Hanning Window.
3. Plot the magnitude response and phase response of the designed filter.
4. Verify the high-pass filtering characteristics of the filter.

Aim (10)	Program (20)	Procedure/Calculations (20)	Results (20)	Record (10)	Viva (20)	Total (100)

Practical Question 2

Design and implement an FIR High-Pass Filter using the **Hanning (Hann) Window** technique in MATLAB for a sampling frequency of **16000 Hz**, cutoff frequency of **4000 Hz**, and filter order of **60**.

Tasks:

1. Design the FIR High-Pass Filter using the specified parameters and Hanning Window.
2. Generate and display the FIR filter coefficients.
3. Obtain the magnitude response and phase response using the freqz() function.
4. Evaluate the passband, stopband, and transition-band characteristics of the filter.

Aim (10)	Program (20)	Procedure/Calculations (20)	Results (20)	Record (10)	Viva (20)	Total (100)

Result

To design and analyze an FIR high-pass filter using a Hanning window.

Student Details

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